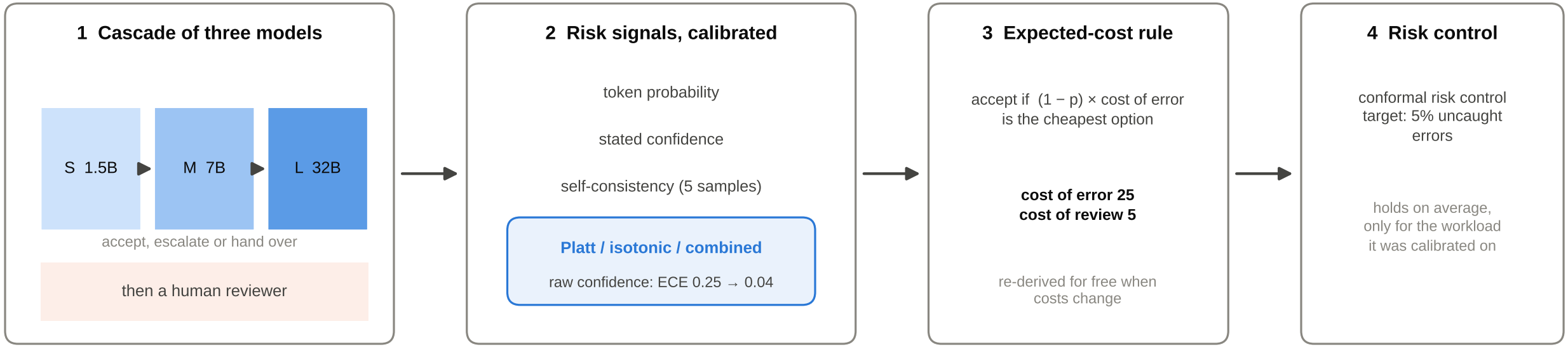
When does calibration change an AI decision? Risk-calibrated, cost-aware LLM cascade with human review



Findings on 2,400 real queries (200 repeated splits)

Calibrated 5.09 vs raw 7.79

cost per query; raw confidence was worse than sending all to a human (5.00)

Tuned thresholds 4.98

calibration adds no savings; stale thresholds cost more in 10 of 12 settings

Guarantee: 4.8% (target 5%)

27% and 47% uncaught errors on two of three workloads calibrated elsewhere